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EXAMINER
BUTLER, MICHAEL E

ART UNIT	PAPER NUMBER
3655	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/093,970	Applicant(s) Fang et al.	
	Examiner MICHAEL E BUTLER	Art Unit 3655	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 23 October 2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-4,6-7 and 9-16 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,6-7,9-12 and 15-16 is/are rejected.

8) ☑ Claim(s) 2-4 and 13-14 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 1/6/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

This action is responsive to the reply filed October 23, 2025. Claims 1-4 and 6-7 and 9-16 are pending. Claims 1, 6-7, 9-12, and 15-16 are rejected. Claims 2-4 and 13-14 are objected to. Claims 5 and 8 are cancelled.

Priority

1. Applicant's claim of priority as a Continuation in Part of 17833614, filed 06/06/2022 which is a Continuation in Part of 17195589 filed 03/08/2022, now U.S. Patent # 11538287 which Claims Priority from Provisional Application 63197021, filed 06/04/2021 and Priority from Provisional Application 62986444, filed 03/06/2020, and Priority from Provisional Application 63024383, filed 05/13/2020, and Priority from Provisional Application 63123531, filed 12/10/2020 is acknowledged.

Claim Rejections - 35 USC § 103

2. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

3. Claim(s) 1, 6, and 9-12 and 15-16 is/are rejected under 35 U.S.C. 103 as being unpatentable over Endo et al. (US20170262277) in view of Ling et al. (US8140358) in further view of Kushwaha et al. (US20200218531) wherein Endo et al. teaches:

(re: cl 1) An apparatus (#100), comprising: a controller that is separate from a vehicle (¶53-55 software management portion #111 of Management server #110 controls software installation onto the vehicle #120 and controls communication inquiry to the permission terminal; ¶62-sends update inquiry to user terminal at one endpoint; "the management server 110 inquires of the portable terminal 140 about whether the update of the software is permitted or not. On this account, when the update permission portion 141 receives, from the management server 110, an inquiry about whether the update is permitted or not, the update permission portion 141 receives an input of the update permission from the user and transmits it to the management server 110"),

comprising: an automation definition circuit structured to interpret a plurality of automation descriptions (¶62-"when the software of the latest version newly stored in the software DB 113 is software that requires a permission of the user, the update permission portion 141 receives an input of an update permission from the user. When the management server 110 is not communicably connected to the vehicle 120, the management server 110 inquires of the portable terminal 140 about whether the update of the software is permitted or not. On this account, when the update permission portion 141 receives, from the management server 110, an inquiry about whether the update is permitted or not, the update permission portion 141 receives

an input of the update permission from the user and transmits it to the management server 110."- interpret operators commands to perform the update)

wherein the automation commands are transmitted from the controller and are received directly by the end points of the plurality of vehicle networks (¶62-sends update inquiry to user terminal at one endpoint; "the management server 110 inquires of the portable terminal 140 about whether the update of the software is permitted or not. On this account, when the update permission portion 141 receives, from the management server 110, an inquiry about whether the update is permitted or not, the update permission portion 141 receives an input of the update permission from the user and transmits it to the management server 110"; ¶69-communication portion #206 of management server #110 communicates to the vehicle #120 to update software or the user terminal #120 to inquire for permission; ¶135-management server transmits software update to the vehicle for installation #S904 fig 9; #140 to #132 fig. 1; ¶63-#140 to #120- the terminal #140 is one endpoint and the automatic driving or driving support system #132 within #120 is at the other endpoint of the network);

and an automation execution circuit structured to provide a plurality of automation commands to end points of the plurality of vehicle networks in response to the automated action plan (¶76, #112 update manager; ¶42-43, Figs. 17-18 - validates and installs software update in the vehicle which is the endpoint of the communication network; ¶73- Interpreter or compiler will interpret the update instructions for installing the new software per the defined action in the computer language; ¶69-communication portion #206 of management server #110 communicates to the vehicle #120 to update software or the user terminal #120 to inquire for permission; ¶135-management server transmits software update to the vehicle for installation #S904 fig 9; ¶53-55

software management portion #111 of Management server #110 controls software installation onto the vehicle #120 and controls communication inquiry to the permission terminal; ¶62-sends update inquiry to user terminal at one endpoint; "the management server 110 inquires of the portable terminal 140 about whether the update of the software is permitted or not. On this account, when the update permission portion 141 receives, from the management server 110, an inquiry about whether the update is permitted or not, the update permission portion 141 receives an input of the update permission from the user and transmits it to the management server 110"; ¶69-communication portion #206 of management server #110 communicates to the vehicle #120 to update software or the user terminal #120 to inquire for permission).

Endo et al. suggests: wherein the automation management circuit is further structured to determine an operational impact for at least one of the automation descriptions (¶187-determinses whether a function subject to the current update has just been performed so as to not duplicated that function), and to provide the automated action plan in response to the operational impact (¶189-if the function has just been performed, restricts it's used until the next ignition cycle).

Ling et al. teaches what Endo et al. lacks of:

(re: cl 1) an automation management circuit structured to provide an automated action plan in response to the plurality of automation descriptions, wherein the automated action plan is structured to provide performance of automated operations for the vehicle without modifying a base software for the vehicle (c29L61-30L5- software updates initialization and defaults and tables but not main program; c32L64-c65L27-remote destination may issue diagnostic

commands; c38L22-25-remote destination may issue diagnostic commands or firmware updates),

and wherein the automated action plan includes vehicle specific information (c29L61-30L5- VIN , year, make, model, transmission ratio, engine; c16L65-17-c17L25-vehicle identification),

including network locations corresponding to respective end points of a plurality of vehicle networks in the vehicle (c30L6-17-known web destinations and server accessed; c32L35-56 user account; c29L61-30L5- VIN , year, make, model, transmission ratio, engine; c16L65-17-c17L25-vehicle identification- the server and web destinations can be an endpoint and the vehicle can be an endpoint:-perform vehicle diagnostic commands or receive the latest software updates or upload data for use in determining insurance rates as taught by Ling et al..)

that are responsive to the plurality of automation descriptions (c36L37-55- transfer data to a destination with a unique identifier-downloads the software from server endpoint to the vehicle endpoint premised upon permission allowed; c32L64-c65L27-remote destination may issue diagnostic commands; c38L22-25-remote destination may issue diagnostic commands or firmware updates).

It would have been obvious before the effective date of the invention to one of ordinary skill in the art for Endo et al. to use an automation management circuit structured to provide an automated action plan in response to the plurality of automation descriptions without modifying the vehicle base software to trigger and perform vehicle diagnostic tests and, trigger vehicle components external the vehicle, and facilitate mechanic specific control of individual components for diagnosis as taught by Ling et al..

It would have been obvious before the effective date of the invention to one of ordinary skill in the art for Endo et al. to have the automated action plan include vehicle specific information so that the firmware update version may be used as to which identifying protocol selections need be used (c27 L17-28-software versions updated via the web) and as one skilled in the art would recognize that knowing the vehicle and configuration can identify which diagnostic tests to perform and what firmware and table calibrations need be updated as taught by Ling et al..

It would have been obvious before the effective date of the invention to one of ordinary skill in the art for Endo et al. to include network locations corresponding to respective end points of a plurality of vehicle networks in the vehicle so as to perform vehicle diagnostic commands or receive the latest software updates or upload data for use in determining insurance rates as taught by Ling et al..

It would have been obvious before the effective date of the invention to one of ordinary skill in the art for Endo et al. to be responsive to the plurality of automation descriptions so that - remote destination may issue diagnostic commands or firmware updates as taught by Ling et al..

Kushwaha et al. teaches any elements Endo et al. lacks including:

and wherein the automation management circuit is further structured to determine an operational impact for at least one of the automation descriptions, and to provide the automated action plan in response to the operational impact (¶43- updates in manner executing fewest steps in update to minimize the impact).

It would have been obvious before the effective date of the invention for one of ordinary skill in the art for Endo et al. to determine an operational impact for at least one of the automation descriptions, and to provide the automated action plan in response to the operational impact to minimize execution disruption time during updates as taught by Kushwaha et al.

Kushwaha et al. teaches what Endo et al. lacks of:

(re: cl 6) wherein the automation management circuit is further structured to deprecate at least one of the plurality of automation descriptions in response to the operations of the automation execution circuit (¶43 removes old software components- Removing old software components reads on deprecate).

It would have been obvious before the effective date of the invention to one of ordinary skill in the art for Endo et al. to deprecate at least one of the plurality of automation descriptions in response to the operations of the automation execution circuit to minimize execution disruption time as taught by Kushwaha et al.

Endo et al. further teaches:

(Re: cl 9) further comprising the vehicle (¶60 #132 executes software installed therein and related to automatic driving support #132 in vehicle fig 1; Function ¶73-executes software related to automatic driving ; ¶77-driving support system and its software installed in a vehicle).

Endo et al. further teaches:

(Re: cl 10) wherein at least one of the plurality of automation descriptions is a user defined automation description (¶285-user permits specific software or denies software and any functions

or disadvantages associated with the denied software component; ¶385- user permits specific software or denies software and any functions or disadvantages associated with the denied software component; abstract-user permits particular software or not).

Endo et al. further teaches:

(Re: cl 11) wherein the automation management circuit is further structured to determine that the user defined automation description is permitted by a user associated with the user defined automation description. (¶62-" when the software of the latest version newly stored in the software DB 113 is software that requires a permission of the user, the update permission portion 141 receives an input of an update permission from the user.")

Endo et al. further teaches:

(Re: cl 12) wherein the user is an operator of the vehicle (¶7-"a permission of a user of the vehicle is required at the time of an update to its latest software."; par54-software related to the automatic driving function or the driving support function is software that requires a permission of a user as a general rule at the time of an update to software of a latest version.").

(re: cl 15) wherein the controller is remote from the vehicle (¶53-55 software management portion #111 of Management server #110 controls software installation onto the vehicle #120 and controls communication inquiry to the permission terminal);

and connected to the end points of the plurality of vehicle networks (¶62-sends update inquiry to user terminal at one endpoint; "the management server 110 inquires of the portable terminal 140 about whether the update of the software is permitted or not."-the permission or not

is responsive; ¶76, #112 update manager; ¶42-43, Figs. 17-18 - validates and installs software update in the vehicle which is the endpoint of the communication network).

via a cloud server (¶62-"when the software of the latest version newly stored in the software DB 113 is software that requires a permission of the user, the update permission portion 141 receives an input of an update permission from the user. When the management server 110 is not communicably connected to the vehicle 120, the management server 110 inquires of the portable terminal 140 about whether the update of the software is permitted or not. On this account, when the update permission portion 141 receives, from the management server 110, an inquiry about whether the update is permitted or not, the update permission portion 141 receives an input of the update permission from the user and transmits it to the management server 110."- interpret operators commands to perform the update; #110 fig. 1).

(re: cl 16) wherein the end points of the plurality of vehicle networks are responsive to the plurality of automation descriptions to perform selected operations (¶62-sends update inquiry to user terminal at one endpoint; "the management server 110 inquires of the portable terminal 140 about whether the update of the software is permitted or not."-the permission or not is responsive; ¶76, #112 update manager; ¶42-43, Figs. 17-18 - validates and installs software update in the vehicle which is the endpoint of the communication network; ¶69-communication portion #206 of management server #110 communicates to the vehicle #120 to update software or the user terminal #120 to inquire for permission; ¶135-management server transmits software update to the vehicle for installation #s904 fig 9; ¶53-55 software management portion #111 of Management server #110 controls software installation onto the vehicle #120 and controls

communication inquiry to the permission terminal; ¶73- Interpreter or compiler will interpret the update instructions for installing the new software per the defined action in the computer language).

4. Claim(s) 7 is/are rejected under 35 U.S.C. 103 as being unpatentable Endo et al. (US20170262277) in view of Ling et al. 8140358 in further view of Kushwaha et al. (US20200218531) in further view of Searle et al.; Endo et al. in view of Ling et al. in further view of Kushwaha et al. teaches the elements previously discussed and Searle et al. teaches what Endo et al. lacks of:

wherein the automation management circuit is further structured to deprecate at least one of the plurality of automation descriptions by performing at least one operation selected from the operations consisting of updating the automated action plan to remove the at least one of the plurality of automation descriptions (¶403-"5 For updates, the DM Client will collect all nodes that have an EXEC operation. These will be the ./DownloadAndUpdate child node of the FUMO object. All FUMO nodes with REMOVE operation identify applications that should be removed from the device. The DM client then generates an update identifier, that ensures that all of the updates received in this sync are applied at the same time, and that the installation events are issued in the correct order");

deleting stored data responsive to the at least one of the plurality of automation descriptions (¶468-"4 The Installer checks to see if any FUMO nodes are ready to be removed by seeing if any node has the Ext/State of READY_TO_REMOVE. The Installer should only continue to perform installation of new applications, or updates if all the nodes that are to be removed have the state of WALKING_DEAD, which means that the custom removal code

has been invoked for all of the updates. The Installer sets the FUMO Ext/State value to be CUSTOM_DEINSTALL_IN_PROGRESS “);

deleting a stored recipe corresponding to the at least one of the plurality of automation descriptions (¶405-“15 Mark the Ext/State to be READY_TO_REMOVE this will allow the nodes that should be removed to be identified later on in the update processing.”);

or adjusting at least one of a priority value or a hierarchy value corresponding to the at least one of the plurality of automation descriptions (¶402-“ The OMA-DM server responds with EXEC, ADD, UPDATE or REMOVE operations to the local tree”).

It would have been obvious before the effective date of the claimed invention for one of ordinary skill in the art for Endo et al. to interpret a priority value corresponding to at least one of the plurality of automation descriptions, and wherein the automation management circuit is further structured to provide the automated action plan in response to the priority value to assure update components are added in a manner in which their base node component from which they depend has been installed first assuring all linked components are not missing dependencies as taught by Searle et al.; OR it would have been obvious before the effective date of the invention to one of ordinary skill in the art for Searle et al. to delete stored data responsive to the at least one of the plurality of automation descriptions to be sure to not remove any nodes that may need be retained due further updates needing their operation or node branch dependency as taught by Searle et al.; OR it would have been obvious before the effective date of the invention to one of ordinary skill in the art for Searle et al. to delete a stored recipe corresponding to the at least one of the plurality of automation descriptions so as to make sure old software not compatible with the updates does not interfere with the new updates due to the old software’s presence as taught by Searle et al.; OR it would have been obvious before the effective date of the invention to one

of ordinary skill in the art for Searle et al. to be adjusting at least one of a priority value or a hierarchy value corresponding to the at least one of the plurality of automation descriptions to be assured the new components are placed based upon the presence of any components they depend upon being in proper place and location first as taught by Searle et al..

Allowable Subject Matter

5. Claims 2-4 and 13-14 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

The following is a statement of reasons for the indication of allowable subject matter: The prior art does not teach the features of described in the rejection to base claim 1 in conjunction with the further limitation of claim 2 of: Searle et al. (US20160294605) wherein Endo et al. teaches the elements previously discussed and Searle et al. teaches what Endo et al. lacks of: (re: cl 2) wherein the automation definition circuit is further structured to interpret a priority value corresponding to at least one of the plurality of automation descriptions, and wherein the automation management circuit is further structured to provide the automated action plan in response to the priority value (¶403-"5 For updates, the DM Client will collect all nodes that have an EXEC operation. These will be the ./DownloadAndUpdate child node of the FUMO object. All FUMO nodes with REMOVE operation identify applications that should be removed from the device. The DM client then generates an update identifier, that ensures that all of the updates received in this sync are applied at the same time, and that the installation events are issued in the correct order"). Of particular interest is Searle et al. (US20160294605) which teaches: wherein Endo et al. teaches the elements previously discussed and Searle et al. teaches

what Endo et al. lacks of: (re: cl 2) wherein the automation definition circuit is further structured to interpret a priority value corresponding to at least one of the plurality of automation descriptions, and wherein the automation management circuit is further structured to provide the automated action plan in response to the priority value (¶403-"5 For updates, the DM Client will collect all nodes that have an EXEC operation. These will be the ./DownloadAndUpdate child node of the FUMO object. All FUMO nodes with REMOVE operation identify applications that should be removed from the device. The DM client then generates an update identifier, that ensures that all of the updates received in this sync are applied at the same time, and that the installation events are issued in the correct order").

The prior art does not teach the features of described in the rejection to base claim 1 in conjunction with the further limitation of claim 3 of: wherein the automation definition circuit is further structured to interpret a hierarchy value corresponding to at least one of the plurality of automation descriptions; and wherein the automation management circuit is further structured to provide the automated action plan in response to the hierarchy value. Of particular interest is Cho et al. (US20200057630) which teaches: teaches what Endo et al. lacks of: (re: cl 3) wherein the automation definition circuit is further structured to interpret a hierarchy value corresponding to at least one of the plurality of automation descriptions (¶87-Table 6-update contingent upon child elements-modules having dependent child elements have a hierarchical relationship between the parent and child elements so the parent may have present the subroutines, functions, and data necessary to execute the functions and subroutines necessary needed for execution of the parent),

and wherein the automation management circuit is further structured to provide the automated action plan in response to the hierarchy value (¶87-Table 6-update contingent upon

child elements-the parent module's execution called upon is further structured to access or execute the child module by having a linked hierarchal relationship). It would not have been obvious to one of ordinary skill in the art to combine Endo et al. in view of Ling et al. in further view of Kushwaha et al. with Cho et al..

The prior art does not teach the features of described in the rejection to base claim 1 in conjunction with the further limitation of claim 4 of: Of particular interest is Maru et al. (US20160294605) which teaches: wherein Endo et al. teaches the elements previously discussed and Maru et al. teaches what Endo et al. lacks of: (re: cl 4) wherein the automation management circuit is further structured to determine a dependency value between at least two of the plurality of automation descriptions, and to provide the automated action plan in response to the dependency value (¶56-determines whether there is a sequence dependency in the update components).

The prior art does not teach the features of described in the rejection to base claim 1 in conjunction with the further limitation of claim 13 of: The update steps are each structured to actuate one or more actuators. Of particular interest is Vangelov et al. (US20190217777) which teaches: The update steps are each structured to actuate one or more actuators (abstract-activates lights, door handles, door handles, audiovision, switches, communications to audiovisual or tactile, electromechanical activation; ¶75-activates horn or audio or voice and speakers chirp horn virtual or tactile). It would not have been obvious to one of ordinary skill in the art to combine Endo et al. in view of Ling et al. in further view of Kushwaha et al. with Vangelov et al..

Response to Argument/Amendments

6. Applicant's amendment was sufficient in overcoming the previous rejection to independent claim 1 evidenced by Endo et al. in view of Ling et al. and dependent claims 9-12 and 15-16. However, Kushwaha et al. teaches any elements Endo et al. lacks including: wherein the automation management circuit is further structured to determine an operational impact for at least one of the automation descriptions, and to provide the automated action plan in response to the operational impact (¶43- updates in manner executing fewest steps in update to minimize the impact). In addition to that-primary reference Endo et al. suggests wherein the automation management circuit is further structured to determine an operational impact for at least one of the automation descriptions (¶187-determinses whether a function subject to the current update has just been performed so as to not duplicated that function), and to provide the automated action plan in response to the operational impact (¶189-if the function has just been performed, restricts it's used until the next ignition cycle). Applicant argues that Kushwaha et al. modifies base software. Secondary reference Ling et al. had been brought in as modifying Endo et al. in that it expressly teaches the negative limitation of: to provide performance of automated operations for the vehicle without modifying a base software for the vehicle (c29L61-30L5- software updates initialization and defaults and tables but not main program; c32L64-c65L27-remote destination may issue diagnostic commands; c38L22-25-remote destination may issue diagnostic commands or firmware updates). In attempting to distinguish over the prior art by claiming what applicant's invention fails to do, applicant forgets that inventions are granted premised upon what they bring to the art, not what they fail to bring. Further, Kushwaha et al. teaches what may be updated can include software components which can include calibrations or offsets or reads on even subroutines and component functions that are not necessarily a part of

the broad term base software or executive portion of the software. Kushwaha et al. teaches his disclosure may be used for a complete software change including the base software or it may be used for altering components and leaving the base software untouched.

Applicant's amendment was effective in overcoming the previous rejection to claims 2-4, and 9-12 and 15-16, but with Kushwaha et al. brought in to base claim 1 and with the elements of claims 9-12 and 15-16 found in primary reference Endo et al., those claim's rejections are evidenced by Endo et al. in view of Ling et al. in further view of Kushwaha et al. as with base claim 1.

Applicant's amendment does not distinguish over dependent claims 6-7 as Kushwaha et al. was previously used as the tertiary reference.

The amendment to base claim 1 was sufficient to bring claims 2-4 and 13-14 to allowability by overcoming the motivation for combining the four references necessary to teach the claimed invention.

Conclusion

7. A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 10/23/2025 has been entered.

8. Any inquiry concerning this communication or earlier communications from the examiner should be directed to MICHAEL E BUTLER whose telephone number is (571)272-6937.

The examiner can normally be reached on Tuesday, Wednesday, or Thursday. Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jacob Scott can be reached on 571-270-3415. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/M.E.B/
Examiner, Art Unit 3655

/JACOB S. SCOTT/
Supervisory Patent Examiner, Art Unit 3655